TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Young’s Modulus of the Material of a Wire by Searle’s Method

Two wires from one support and a micrometer between their ends — measuring the extension of one wire against the other, so support and temperature errors cancel.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 4 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find Young’s modulus Y of a wire by Searle’s apparatus. Two wires of the same material hang side by side from the **same rigid support**: the **reference wire** carries a fixed load (just enough to keep it taut) and the **experimental wire** carries the variable load. Their lower ends are joined by a horizontal frame carrying a spirit level and a **micrometer screw**. When the loaded wire stretches, the frame tilts; the micrometer is turned to restore the spirit level to horizontal, and the micrometer’s movement *is* the extension ΔL — measured **relative to the reference wire**, so any yielding of the support or thermal expansion common to both wires cancels. Then

$$Y = F\,L / (A\,\Delta L) = 4\,M\,g\,L / (\pi\,d^2\,\Delta L)$$

exactly as in the vernier method (Experiment 3); only the extension measurement is cleverer — a micrometer reads to 0.01 mm, ten times finer than the vernier arrangement.

2. Apparatus and procedure

- **Apparatus:** Searle’s apparatus (two wires, frame, spirit level, micrometer), slotted masses and two hangers, screw gauge, metre rule.
- Measure L of the experimental wire and its diameter d at several places (mean of perpendicular pairs).
- Put equal dead loads on both hangers; adjust the micrometer until the spirit-level bubble is centred; record the micrometer reading.
- Add load to the experimental wire only, in equal steps; after each step, restore the bubble to centre with the micrometer and record the new reading. The difference from the initial reading is ΔL for that load.

- Unload in steps, repeating the readings (loading/unloading agreement checks elasticity). Plot ΔL against load; from the slope compute Y.

3. Worked calculation (model data)

RESULT

Experimental wire: L = 1.50 m; d = 0.40 mm → A = $\pi(2.0 \times 10^{-4})^2 = 1.257 \times 10^{-7}$ m².
From the graph: a load increase of 3.0 kg (F = 29.4 N) gives micrometer movement ΔL = 1.75 mm = 1.75×10^{-3} m.
Y = $(29.4 \times 1.50)/(1.257 \times 10^{-7} \times 1.75 \times 10^{-3}) = 44.1/(2.20 \times 10^{-10})$ = **2.0×10^{11} Pa** — steel, in agreement with the vernier method's result for the same material, obtained here with a shorter wire because the micrometer is finer.

4. Precautions

- Always approach the bubble-centred position turning the micrometer the **same way** (backlash in the screw otherwise shifts readings).
- The two wires must hang freely without touching each other or the frame; a rubbing wire falsifies the level.
- Keep loads within the elastic limit and add them gently; wait for the bubble to settle before reading.
- The reference wire's load stays constant throughout — it is the experiment's fixed point.
- Because ΔL is small, avoid breathing warmth and hand heat on the wires; wait after handling.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. What does the micrometer actually measure in Searle's method, and why is that better than measuring the wire's end position directly?
2. A wire (L = 2.00 m, d = 0.50 mm) extends 1.25 mm under 2.0 kg in Searle's apparatus. Compute Y.
3. Why must the micrometer always be turned in the same sense when centring the bubble?
4. Both wires are of the same material and length. Why does that matter for temperature compensation?
5. What goes wrong if the load on the reference wire is changed mid-experiment?
6. Compare Searle's and the vernier method: one advantage and one limitation of each.

Answer key

1. It measures the extension of the experimental wire *relative to the reference wire* — the extra length the loaded wire gains over its unloaded twin. Support sag and thermal expansion move both wires together and cancel, so the small true extension survives measurement instead of drowning in apparatus drift.
2. $A = \pi(2.5 \times 10^{-4})^2 = 1.963 \times 10^{-7}$ m²; F = 19.6 N; $Y = (19.6 \times 2.00)/(1.963 \times 10^{-7} \times 1.25 \times 10^{-3}) = 39.2/(2.454 \times 10^{-10}) = 1.6 \times 10^{11}$ Pa — noticeably below steel's 2.0×10^{11} Pa, so this wire is a different metal; the method reports the material as it finds it.
3. Micrometer screws have backlash — a small dead travel when the turning direction reverses. Approaching the final setting always from the same side keeps the screw's contact on the same thread flank, so readings are consistent.
4. Thermal expansion then lengthens both wires by the same amount for a given temperature drift, and the relative measurement is unaffected. Different materials (different expansion coefficients) would leave a residual drift in the difference.
5. The reference wire's length changes with its load, so the "fixed" reference moves: every subsequent ΔL is measured against a shifted zero, corrupting the whole run. Its load is part of the apparatus setting, not a variable.
6. Searle's: finer resolution (0.01 mm micrometer) and built-in compensation, but a delicate level-and-frame arrangement disturbed by vibration and touching. Vernier: simpler and robust, but coarser readings and compensation only through its dummy wire. Both live or die by the diameter measurement ($Y \propto 1/d^2$).
